由于 $a_1 = a_2 = z > 0$，所以 ReLu 导数为 1，所以

$$\frac{\partial L}{\partial a_1} = \frac{349}{2400} \qquad \frac{\partial L}{\partial a_2} = -\frac{173}{2400}$$

分别记为 $\bar{a}_1$ $\bar{a}_2$

由
$$\frac{\partial a_j}{\partial q_t} = S_{j+t-1}$$

$$\frac{\partial L}{\partial q_t} = \sum_{j=1}^{2} \frac{\partial L}{\partial a_j} \frac{\partial a_j}{\partial q_t}$$

得
$$\frac{\partial L}{\partial q_t} = \bar{a}_1 S_t + \bar{a}_2 S_{t+1}$$

所以 $\frac{\partial L}{\partial q_1} = \frac{525}{2400}$ $\frac{\partial L}{\partial q_2} = \frac{522}{2400}$ $\frac{\partial L}{\partial q_3} = \frac{-868}{2400}$

所以 $\nabla_q L = \frac{1}{2400}\begin{bmatrix} 525 \\ 522 \\ -868 \end{bmatrix}$

对 S_2 的梯度 $\frac{\partial L}{\partial S_2} = \bar{a}_1 q_2 + \bar{a}_2 q_1$

$$= \frac{1}{1600}$$

卷积核更新

$$q^{new} = q^{old} - \eta \nabla_q L$$

$$= \begin{bmatrix} 1 \\ 0.5 \\ 0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 7/32 \\ 87/400 \\ -217/600 \end{bmatrix}$$

$$= \begin{bmatrix} 313/320 \\ 1913/14000 \\ 3217/6000 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9781 \\ 0.4783 \\ 0.536167 \end{bmatrix}$$